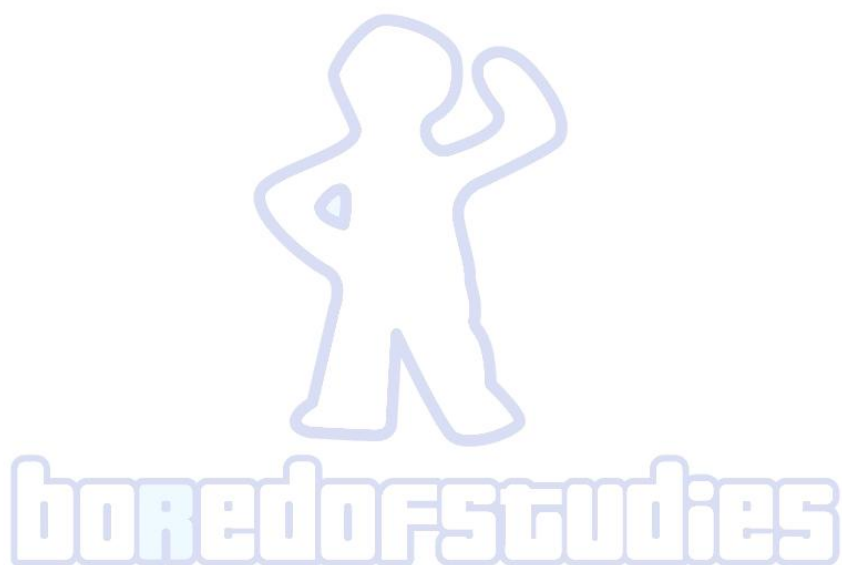




2017 Bored of Studies Trial Examinations

Mathematics Extension 2

9th October 2017

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A reference sheet has been provided.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 1 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 21

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

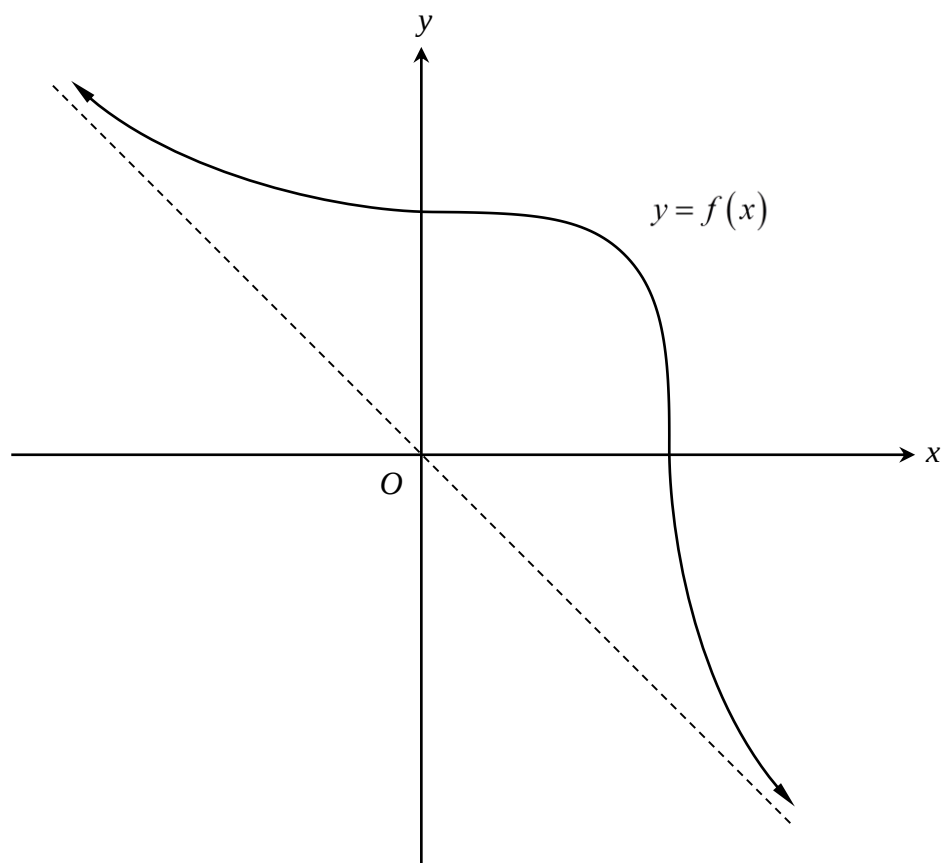
Total marks – 10

Attempt Questions 1 – 10

All questions are of equal value

Shade your answers in the appropriate box on the Multiple Choice answer sheet provided.

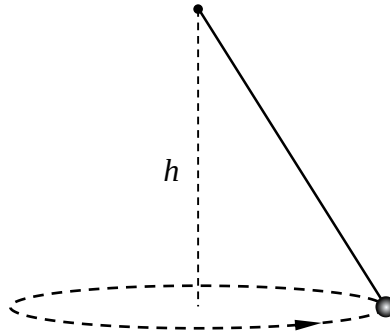
- 1 The diagram below shows a sketch of the function $y = f(x)$.



Which of the following is the most suitable expression for $\frac{dy}{dx}$?

- (A) $\frac{x}{y}$.
- (B) $-\frac{x}{y}$.
- (C) $\frac{x^2}{y^2}$.
- (D) $-\frac{x^2}{y^2}$.

- 2 An object of mass m is attached to a fixed point by a taut inelastic string. It rotates in a circle with constant angular velocity ω , tracing out a cone with the string. Let h be the vertical distance between the particle and the fixed point, as shown in the diagram below.



When the angular velocity is doubled, the value of h becomes $k \times h$ for some constant k .

Which of the following is the value of k ?

- (A) $\frac{1}{4}$.
- (B) $\frac{1}{2}$.
- (C) 2.
- (D) 4.
- 3 For what values of k does the locus of $|z+1|+|z-1|=k$ describe an ellipse?
- (A) $k < 2$.
- (B) $k \leq 2$.
- (C) $k > 2$.
- (D) $k \geq 2$.

- 4 What is the value of the obtuse angle between the asymptotes of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ where } a \text{ and } b \text{ are positive values and } a > b?$$

- (A) $\tan^{-1}\left(\frac{b}{a}\right).$
- (B) $\tan^{-1}\left(\frac{a}{b}\right).$
- (C) $2 \tan^{-1}\left(\frac{b}{a}\right).$
- (D) $2 \tan^{-1}\left(\frac{a}{b}\right).$

- 5 Let $P(z)$ be a polynomial with real coefficients such that $P(\alpha) = \beta$, where α and β are *non-real* numbers.

Which of the following is always true?

- (A) When $P(z)$ is divided by $(z + \alpha)$, the remainder is $-\beta$.
- (B) When $P(z)$ is divided by $(z + \bar{\alpha})$, the remainder is $-\bar{\beta}$.
- (C) When $P(z)$ is divided by $(z - \bar{\alpha})$, the remainder is $\bar{\beta}$.
- (D) The factor $(z - \alpha)$ is the only one that will yield a remainder of β .

- 6 The integral $\int_0^1 \frac{3x^2 + 3x + 2}{(x^2 + 1)(x + 1)} dx$ can be expressed in the form

$$\ln 2 + \int_0^1 \frac{f(x)}{(x^2 + 1)(x + 1)} dx.$$

Which of the following is a possible expression for $f(x)$?

- (A) $x^2 + x + 2$
- (B) $3x^2 - 1$
- (C) $3x^2 + x$
- (D) $4x^2 - 3x + 4$
- 7 Let $w = \frac{az + bz^{-1}}{bz + az^{-1}}$, where $|z| = 1$. What is the value of $|w|$?

- (A) 1.
- (B) $\frac{a}{b}$.
- (C) $\frac{b}{a}$.
- (D) It depends on $\arg(z)$.

8 Suppose $(\sqrt{3}-i)^n$ is a real number. Which of the following is not always an integer?

- (A) n .
- (B) $n/2$.
- (C) $n/3$.
- (D) $n/4$.

9 The graph of $y = \frac{x^3+2}{x}$ has a stationary point at $(1,3)$. Let $P(x) = x^3 - kx + 2$.

For what values of k will $P(x)$ have 3 distinct real roots?

- (A) $k < 0$.
- (B) $0 < k < 1$.
- (C) $1 < k < 3$.
- (D) $k > 3$.

10 Let z be a variable complex number with unit modulus, other than $\pm i$ and ± 1 .

Which of the following expressions has the locus equation $x = 0$?

- (A) $\frac{z+1}{z-1}$.
- (B) $\frac{z-i}{z+1}$.
- (C) $\frac{z+i}{z-1}$.
- (D) $\frac{z-i}{z-1}$.

Total marks – 90

Attempt Questions 11 – 16

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Find 2

$$\int \frac{dx}{\sin x \sqrt{\sin 2x}}.$$

(b) Consider the polynomial $P(z) = z^{n+1} - 1$. Let ω be any root of $P(z)$.

(i) Show that any integer power of ω is a root of the polynomial. 1

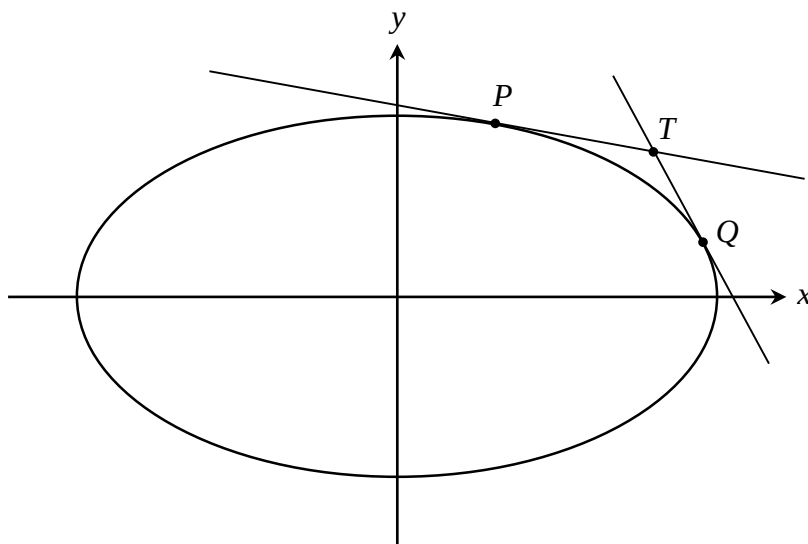
(ii) By considering $P\left(1 - \frac{1}{z}\right)$, or otherwise, show that 2

$$\frac{1}{1-\omega} + \frac{1}{1-\omega^2} + \frac{1}{1-\omega^3} + \cdots + \frac{1}{1-\omega^n} = \frac{n}{2}.$$

Question 11 continues on page 7

Question 11 (continued)

- (c) The diagram below shows two points $P(a \cos \theta, b \sin \theta)$ and $Q(a \sin \theta, b \cos \theta)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. From P and Q , tangents are drawn to intersect at T .



The equation of the tangent drawn from P is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (\text{Do NOT prove this})$$

- (i) Show that T has coordinates

2

$$\left(\frac{a}{\sin \theta + \cos \theta}, \frac{b}{\sin \theta + \cos \theta} \right).$$

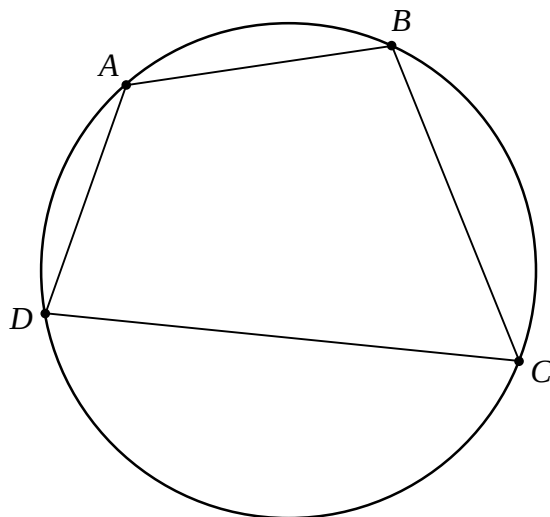
- (ii) Hence, find the locus of T , stating restrictions.

2

Question 11 continues on page 8

Question 11 (continued)

- (d) Let A, B, C and D be concyclic points on the Argand diagram, representing complex numbers a, b, c and d respectively as shown in the diagram below.



- (i) Show that 2

$$|(a-b)(c-d)| + |(a-d)(b-c)| \geq |(a-c)(b-d)|.$$

- (ii) Show that $\frac{(a-b)(c-d)}{(a-d)(b-c)}$ is real. 2

- (iii) Let A, B, C and D be points on the Argand diagram representing complex numbers a, b, c and d respectively. 2

Show that $AB \times CD + BC \times AD = AC \times BD$.

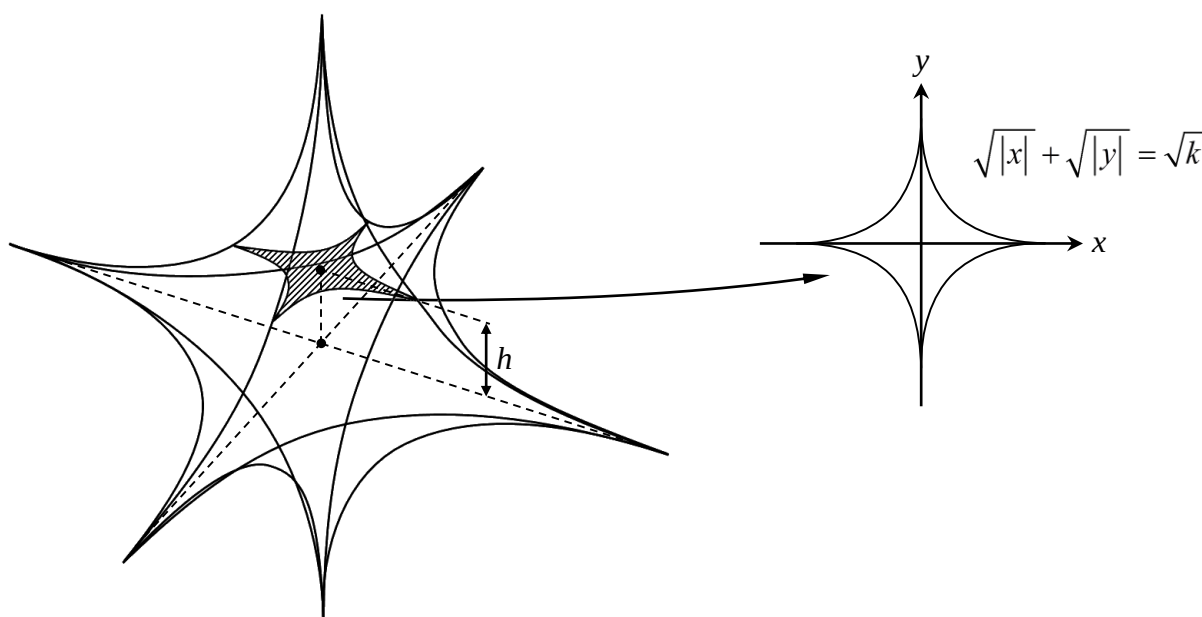
End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) A solid has twelve curved edges, each following the shape of the curve $\sqrt{x} + \sqrt{y} = 1$.

Cross sections taken perpendicular to the vertical axis of symmetry satisfy the relation

$$\sqrt{|x|} + \sqrt{|y|} = \sqrt{k}.$$



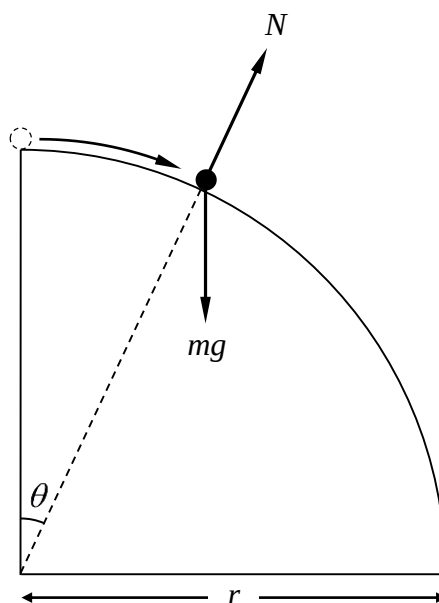
- (i) Show that $\sqrt{k} = 1 - \sqrt{h}$. 1
- (ii) Hence, find the volume of the solid. 2
- (b) (i) Sketch the graph of $y = \frac{x}{x^2 - 1}$. 2
- (ii) Hence, or otherwise, sketch the graph of $y = \frac{e^x}{e^{2x} - 1}$. 2

Question 12 continues on page 10

Question 12 (continued)

- (c) A marble of mass m is initially at rest on top of a smooth circular path with radius r . It is given a push with a negligible amount of force to initiate the movement along the circular path.

Let N be the normal force of the circular path acting on the marble, and g be the magnitude of acceleration due to gravity. Let the angular displacement of the particle be θ , as shown in the diagram below.



- (i) By resolving the forces in the tangential and normal direction of the marble, or otherwise, show that 2

$$N = mg(3\cos\theta - 2).$$

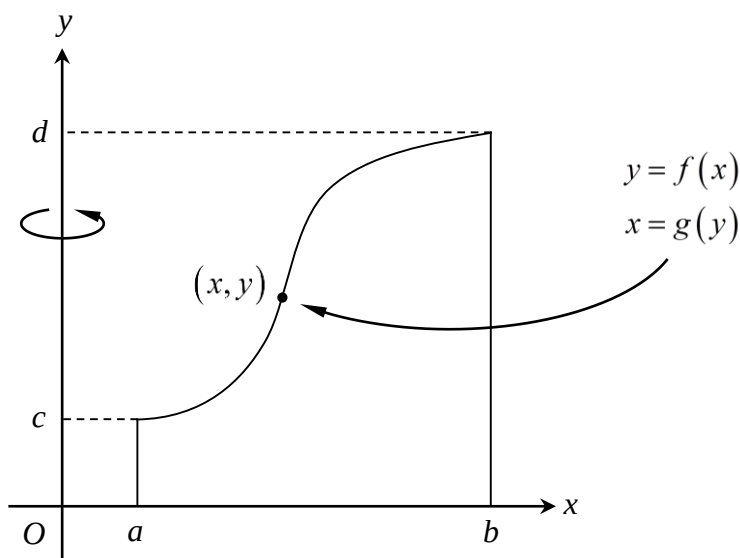
- (ii) When $\theta = \alpha$, the marble leaves the circular surface. 1

Find the exact value of α .

Question 12 continues on page 11

Question 12 (continued)

- (d) The diagram below shows the graph of $f(x)$, which is increasing in the domain $a \leq x \leq b$. Let the inverse function be $x = g(y)$.



The region bound by $y = f(x)$ and the x axis in the domain $a \leq x \leq b$ is rotated about the y axis, forming a solid of revolution.

- (i) Use the slicing method to show that the volume of the solid is 2

$$V = \pi(b^2d - a^2c) - \pi \int_c^d g^2(y) dy.$$

- (ii) Use the substitution $y = f(x)$ and integration by parts to show that 2
the expression in part (i) is equivalent to

$$2\pi \int_a^b x f(x) dx.$$

- (iii) Explain the significance of the result in part (ii). 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Harin enters a casino with \$ k , where k is an integer, and goes to the roulette wheel. If the ball lands on red, which has probability $0 < p < 1/2$, then he wins \$1, otherwise he loses \$1. Harin has no self-control, so he continues until he leaves either disappointed or satisfied. If he loses all \$ k , he leaves the casino disappointed, but if he obtains a total of \$ K , including the money he came in with where $K \geq k$, then he leaves satisfied.

Let P_k be the probability that he eventually leaves disappointed after starting with \$ k .

- (i) Explain why $P_k = pP_{k+1} + (1-p)P_{k-1}$. 1

- (ii) Let $\alpha = \frac{1-p}{p}$ where $\alpha \neq 1$. 1

Show that $P_{k+1} - P_k = \alpha(P_k - P_{k-1})$.

- (iii) Hence show that $P_{k+1} - P_k = \alpha^k(P_1 - 1)$. 1

- (iv) Deduce that $P_k = 1 + (P_1 - 1)\left(\frac{\alpha^k - 1}{\alpha - 1}\right)$. 1

- (v) By considering a similar expression for P_K and then eliminating P_1 , show that 2

$$P_k = \frac{\alpha^K - \alpha^k}{\alpha^K - 1}.$$

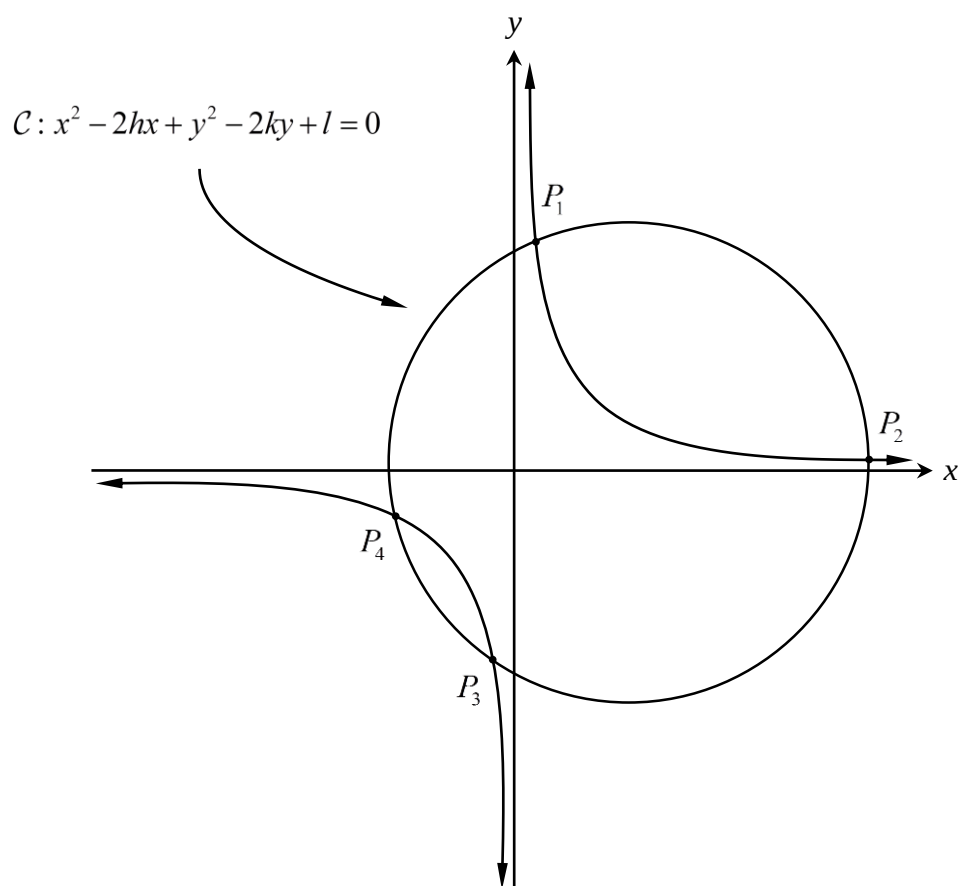
- (vi) Find the probability of leaving satisfied as $K \rightarrow \infty$. 2

Explain the significance of your result.

Question 13 continues on page 13

Question 13 (continued)

- (b) The diagram below shows a circle $C: x^2 - 2hx + y^2 - 2ky + l = 0$ intersecting the hyperbola $xy = c^2$ at four points $P_i \left(cp_i, \frac{c}{p_i} \right)$, where $i = 1, 2, 3, 4$. 3



The *mean point* of n points (x_i, y_i) , where $i = 1, 2, 3, \dots, n$, is defined to be

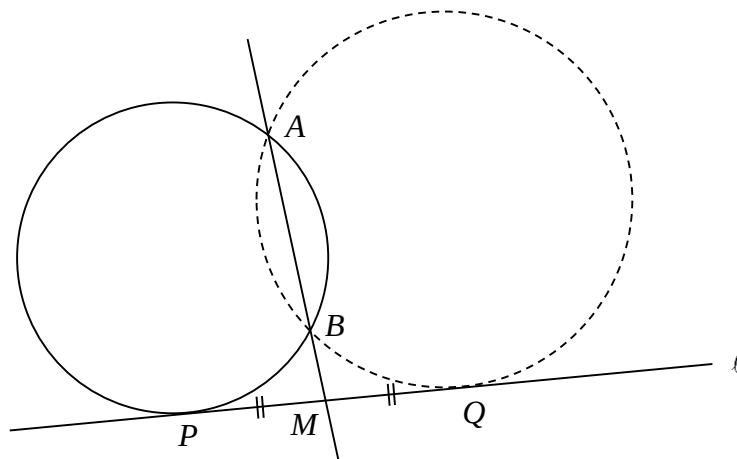
$$M \left(\frac{1}{n} \sum_{i=1}^n x_i, \frac{1}{n} \sum_{i=1}^n y_i \right).$$

Show that the origin, the centre of C and the mean point of P_1, P_2, P_3 and P_4 are collinear.

Question 13 continues on page 14

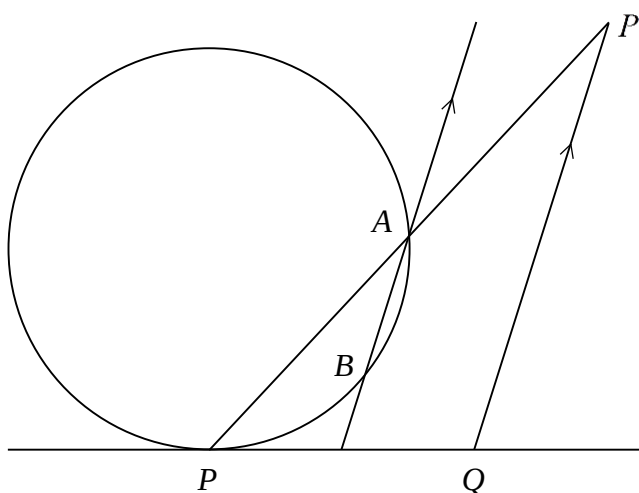
Question 13 (continued)

- (c) Let M be intersection of a tangent ℓ and secant MBA , as shown in the diagram below. Let Q be the point on ℓ such that M is the midpoint of PQ . 2



Show that the circle passing through A , B and Q is tangential to ℓ .

- (d) Let PQ be a tangent to a circle at P . Let A be any point on the circle and P' be on PA produced so that $AP = AP'$, as shown in the diagram below. Let B be the point on the circle such that $AB \parallel P'Q$. 2



Use part (c) to show that $\angle BQP = \angle AQP'$.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Two players A and B compete against each other in a coin-tossing game. If the coin lands on heads, then A scores a point and if the coin lands on tails, then B scores a point. The first person to have a lead of two points wins the game. If nobody has a lead of two points after the $2n^{\text{th}}$ toss is made, then the game is tied. 2

How many possible ways can the games be played?

- (b) Let a_k and b_k be real for $k = 1, 2, \dots, n$. 3

Use mathematical induction to prove that

$$(a_1b_1 + a_2b_2 + \dots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2).$$

- (c) Consider the functions $f(x) = x - \frac{a^2}{x}$ and $g(x) = 2a \ln\left(\frac{x}{a}\right)$.

- (i) Show that $f'(x) \geq g'(x)$ when $x \geq a$. 1
- (ii) Hence, explain why $f(x) \geq g(x)$ when $x \geq a$. 1
- (iii) Sketch the graph of $y = f(x) - g(x)$. 1

Question 14 continues on page 16

- (d) An object of unit mass is projected vertically from the origin with initial speed u . It experiences resistance force of magnitude kv , where k is a positive constant and v is the speed of the object. Let $x(t)$ be the height of the object from the ground at time t seconds, and let g be acceleration due to gravity. After attaining a maximum height, the object falls towards the ground and experiences a resistance force of the same magnitude as the upwards journey.

- (i) Show that 3

$$x(t) = \frac{1}{k^2}(g + ku)(1 - e^{-kt}) - \frac{gt}{k}.$$

- (ii) Find the time T taken for the object to reach its maximum height. 1

- (iii) Show that when $t = 2T$ 2

$$x(2T) = \frac{1}{k^2} \left[g + ku - \frac{g^2}{g + ku} - 2g \ln \left(\frac{g + ku}{g} \right) \right].$$

- (iv) Hence, using the result in part (c), or otherwise, explain why the object takes longer on the downwards journey than the upwards journey. 1

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Show that $\sin x \leq x \leq \cos x$ for $0 \leq x \leq \frac{\pi}{6}$. **3**

(b) Suppose that

$$I_n = \int_0^{\frac{\pi}{6}} \frac{\sin^{2n} x}{\cos x} dx,$$

for some integer $n \geq 0$.

(i) Show that **3**

$$I_n = \frac{\ln 3}{2} - \sum_{k=1}^n \frac{1}{(2k-1)2^{2k-1}}.$$

(ii) Using the result in part (a), or otherwise, show that **2**

$$0 \leq I_n \leq \frac{1}{2n} \left(\frac{\pi}{6} \right)^{2n}.$$

(iii) Hence, find the limiting value of **1**

$$1 + \frac{1}{3 \times 2^2} + \frac{1}{5 \times 2^4} + \frac{1}{7 \times 2^6} + \dots$$

Question 15 continues on page 18

Question 15 (continued)

- (c) Suppose the polynomial $P(z) = z^3 + \beta z^2 + k$ has two non-real roots α and $\bar{\alpha}$, where $\alpha = x + iy$, and a real root β .

(i) Show that $\beta = -x$. 1

(ii) Show that $|\alpha|^2 = 2\beta^2$. 2

(iii) Show that α and $\bar{\alpha}$ lie on the lines $y = \pm x$ in the complex plane. 1

(iv) The roots of $P(z)$ form a triangle in the complex plane. 2

Show that the area of the triangle is $|\alpha|^2$.

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

A *prime number* is a positive integer divisible only by 1 and itself. Let p be a prime number that is larger than another fixed positive integer n .

Let $Q(x)$ be a polynomial of degree r , with equation

$$Q(x) = \frac{x^{p-1}(1-x)^p(2-x)^p(3-x)^p \dots (n-x)^p}{(p-1)!}.$$

Define

$$T(x) = Q(x) + Q^{(1)}(x) + Q^{(2)}(x) + \dots + Q^{(r)}(x),$$

where $Q^{(i)}(x) = \frac{d^i}{dx^i}[Q(x)]$, that is the i -th derivative of $Q(x)$ for $i = 1, 2, \dots, r$.

- (a) Since $Q(x)$ is a polynomial of degree r with lowest-degree term x^{p-1} , it may be expressed as 1

$$Q(x) = a_r x^r + a_{r-1} x^{r-1} + \dots + a_{p-1} x^{p-1},$$

for some integers $a_{p-1}, a_p, \dots, a_r$.

Explain why for any integer $i \geq p$ that

$$Q^{(i)}(x) = pR_i(x),$$

where $R_i(x)$ is some polynomial with integer coefficients.

- (b) Show that $Q^{(p-1)}(0) = (n!)^p$. 2

- (c) Hence, explain why $\frac{Q^{(p-1)}(0)}{p}$ is not an integer. 1

Question 16 continues on page 20

Question 16 (continued)

- (d) Explain why for any integer $1 \leq m \leq n$, 1

$$T(m) = Q^{(p)}(m) + Q^{(p+1)}(m) + Q^{(p+2)}(m) + \dots + Q^{(r)}(m),$$

and

$$T(0) = Q^{(p-1)}(0) + Q^{(p)}(0) + Q^{(p+1)}(0) + \dots + Q^{(r)}(0).$$

- (e) Let $c_0, c_1, \dots, c_n$ be any set of integers such that $0 < |c_0| < p$. 2

Explain why

$$\frac{c_0 T(0) + c_1 T(1) + c_2 T(2) + \dots + c_n T(n)}{p},$$

is not an integer.

- (f) Consider the function $f(x) = e^{-x} T(x)$. 1

Show that $f'(x) = -e^{-x} Q(x)$.

- (g) For any real values $a < b$, there exists some $0 < \alpha < 1$ such that 1

$$f'(a + \alpha(b-a)) = \frac{f(b) - f(a)}{b-a} \quad \text{(Do NOT prove this)}$$

Use this result to show that for any, $1 \leq k \leq n$, there exists $0 < \alpha < 1$ such that

$$T(k) - e^k T(0) = -k e^{k(1-\alpha)} Q(k\alpha).$$

Question 16 continues on page 21

Question 16 (continued)

(h) Define $\beta_k = -ke^{k(1-\alpha)}Q(k\alpha)$. Show that $|\beta_k| \leq \frac{e^n n^p (n!)^p}{(p-1)!}$. 2

(i) Let $P(x) = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$ be a polynomial with integer coefficients such that $0 < |c_0| < p$. 2

Suppose that $P(e) = 0$, show that

$$c_0T(0) + c_1T(1) + c_2T(2) + \dots + c_nT(n) = c_1\beta_1 + c_2\beta_2 + c_3\beta_3 + \dots + c_n\beta_n.$$

(j) It can be shown that if y is fixed, then $\frac{y^p}{p!} \rightarrow 0$ as p gets large. 1

Explain why for each k , there exists some p such that $|c_k\beta_k| < \frac{1}{n}$.

(k) Hence, prove that e cannot be a root of any polynomial with integer coefficients. 1

End of Exam